



The Orto Botanico di Padova (Padua Botanical Garden), the oldest existing botanical garden in Europe, continues to be a major center for research in botany and pharmacology even today.

GERMAN CLERGYMAN AND INSTRUMENT-MAKER GEORG HARTMANN (1489–1564), was the first to notice and describe, in 1544, the phenomenon of magnetic inclination. Also known as **magnetic dip**, this is the phenomenon whereby the needle of a compass follows the line of Earth's magnetic field, which curves around Earth's surface. As a result, the north-pointing end of a compass needle tends to point slightly downward in the Northern Hemisphere and upward in the Southern Hemisphere. Hartmann's discovery was not widely known until centuries later. In 1581, English instrument-maker Robert Norman published his own account of the phenomenon.

“ MATHEMATICS IS... ITS OWN EXPLANATION... FOR THE **RECOGNITION** THAT A **FACT** IS SO, IS THE **CAUSE UPON WHICH WE BASE THE PROOF.** ”

Gerolamo Cardano, Italian mathematician, in *De Vita Propria Liber*

MICHAEL SERVETUS (1511–53)

Spanish scientist Miguel Servet, also known as Michael Servetus, wrote several treatises on medicine and human anatomy. He was the first European to correctly explain pulmonary circulation in *Christianismi Restitutio* (*The Restoration of Christianity*). His theological works were considered heretical, and he was burned at the stake in Geneva for his views.



In 1545, Italian mathematician **Girolamo Cardano** (1501–76) published *Ars Magna* (*The Great Art*), an important book on algebra. He presented **solutions to cubic and quartic equations**, involving unknown quantities to the power of three and four respectively. He drew on his own ideas as well as those of compatriots such as **Niccolò Fontana Tartaglia** (1499–1557), who translated works of Euclid and Archimedes. *Ars Magna* made reference for the first time to **imaginary numbers**—those numbers that are a multiple of the square root of -1 .

Italy was also becoming a center for botanical research, with the opening of the **botanical garden** in Padua, in 1545. The

first of its kind and a model for subsequent botanical gardens, it was established by the Senate of the Venetian Republic. It was used for **growing and studying medicinal plants** and comprised a circular plot of land, symbolizing the world, surrounded by water. The **first custodian** of the gardens was **Luigi Squalermo** (1512–70), also known as Anguillara. He cultivated around **1,800 species of medicinal herbs**, making a significant contribution to the modern scientific studies of botany, medicine, and pharmacology.

“ **CLOTHES, LINEN, ETC...** NOT THEMSELVES CORRUPT, **CAN... FOSTER THE ESSENTIAL SEEDS OF THE CONTAGION** AND THUS CAUSE INFECTION. ”

Girolamo Fracastoro, Italian physician, poet, and geologist, 1546

PHYSICIAN, GEOLOGIST, AND POET GIROLAMO FRACASTORO (1478–1553) published his most important work, *On Contagion and Contagious Diseases*, in Italy in 1546. Best known at the time for his poem *Syphilis, or the French Disease* in 1530, he covered his examination of diseases in greater depth in *On Contagion*, offering an early **explanation of the mechanism by which diseases are spread**. His theory was that each disease is caused by very small bodies, or “spores,” which are carried in the body, skin, and clothing of the person affected. These minute bodies, he believed, could multiply rapidly, and be transmitted from person to person by physical contact by handling unwashed clothes, or

3.4 BILLION YEARS THE AGE OF THE EARLIEST FOSSILS, OF SINGLE CELLS, DISCOVERED IN AUSTRALIA

even through the air. Despite being initially accepted by the medical establishment, his ideas had little effect on the treatment and prevention of disease until his theory was proven right by Louis Pasteur (see 1857–58) and others centuries later. Fracastoro also took an interest in the



Ammonite fossil
Early scholars believed that fossils were the remains of animals laid down in the biblical flood. By the 16th century, people were considering other theories.

1544 Georg Hartmann discovers magnetic dip

1545 The botanical garden in Padua is opened

1545 Gerolamo Cardano publishes his treatise on algebra, *Ars magna*

1546 Girolamo Fracastoro publishes *On Contagion and Contagious Diseases*

1546 In a letter to a colleague, Gerardus Mercator mentions that magnetic and geographic North poles are different

1546 Georgius Agricola publishes *On the Nature of Fossils*